Actionable Real-Time Modeling of Surgical Team Dynamics via Time-Expanded Interaction Graphs

Vincenzo Marco De Luca[1], Antonio Longa [2], Giovanna Varni[1], Andrea Passerini [1]

[1]University of Trento , [2]UiT The Arctic University of Norway

Operating Rooms Design

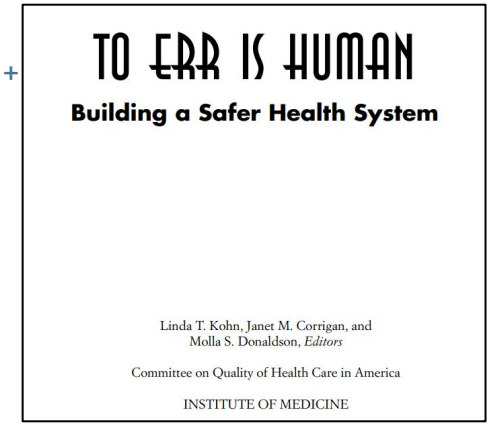


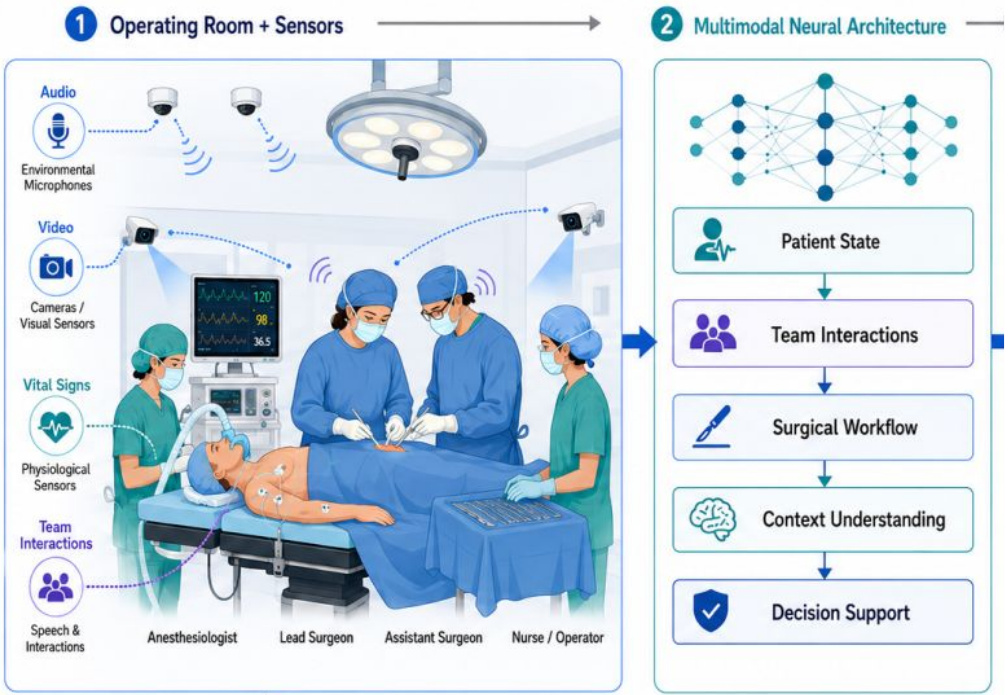
Figure: (Brown, 2001)

+ Medical errors are the **third leading cause of death** in the US
+ One of the main reasons for medical errors is **poor teamwork in the OR**
+ Lack of datasets capturing interactions in the OR
+ Lack of proper data annotations for team data
+ Lack of architectures modeling team behavior generating accurate predictions
+ Design **agents able to enhance Teamwork, Team efficiency, and Team-Decision Making**

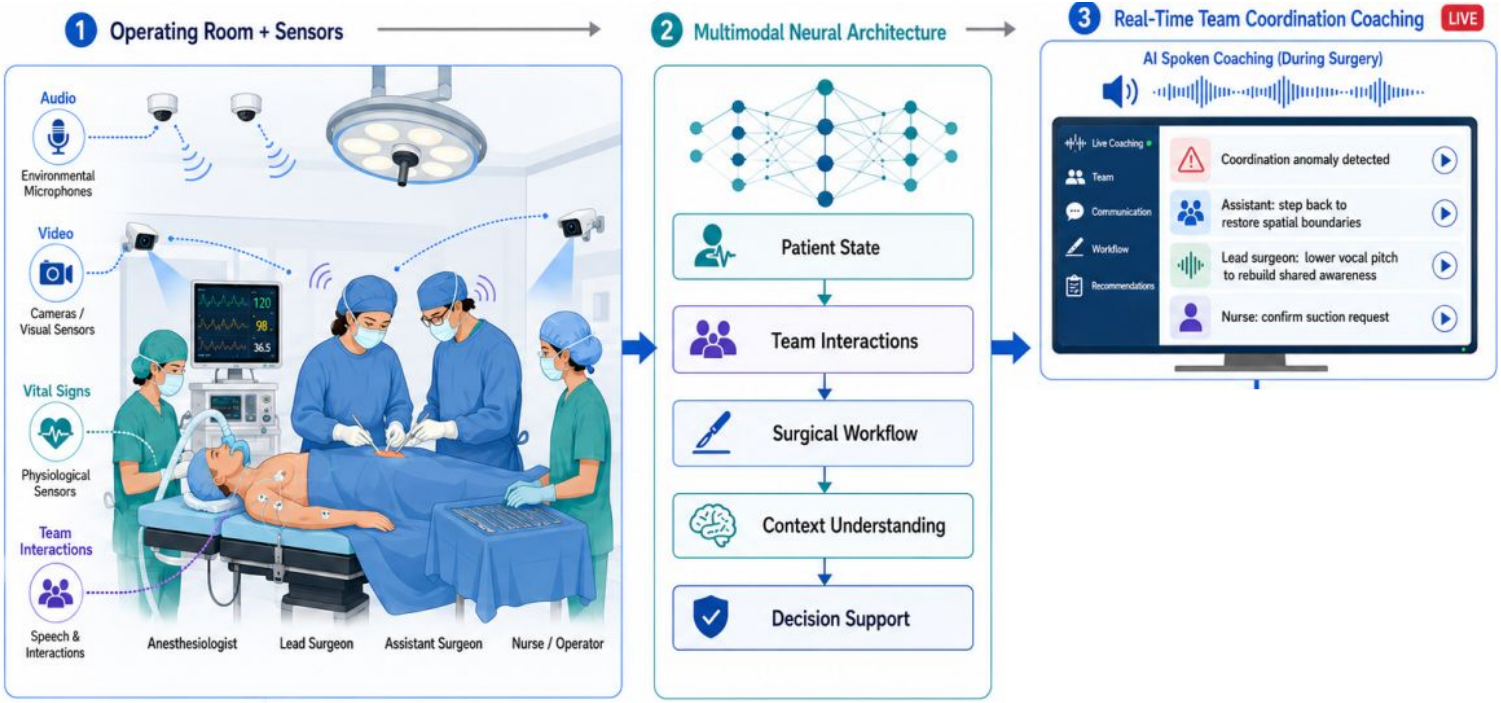
Operating Rooms Design



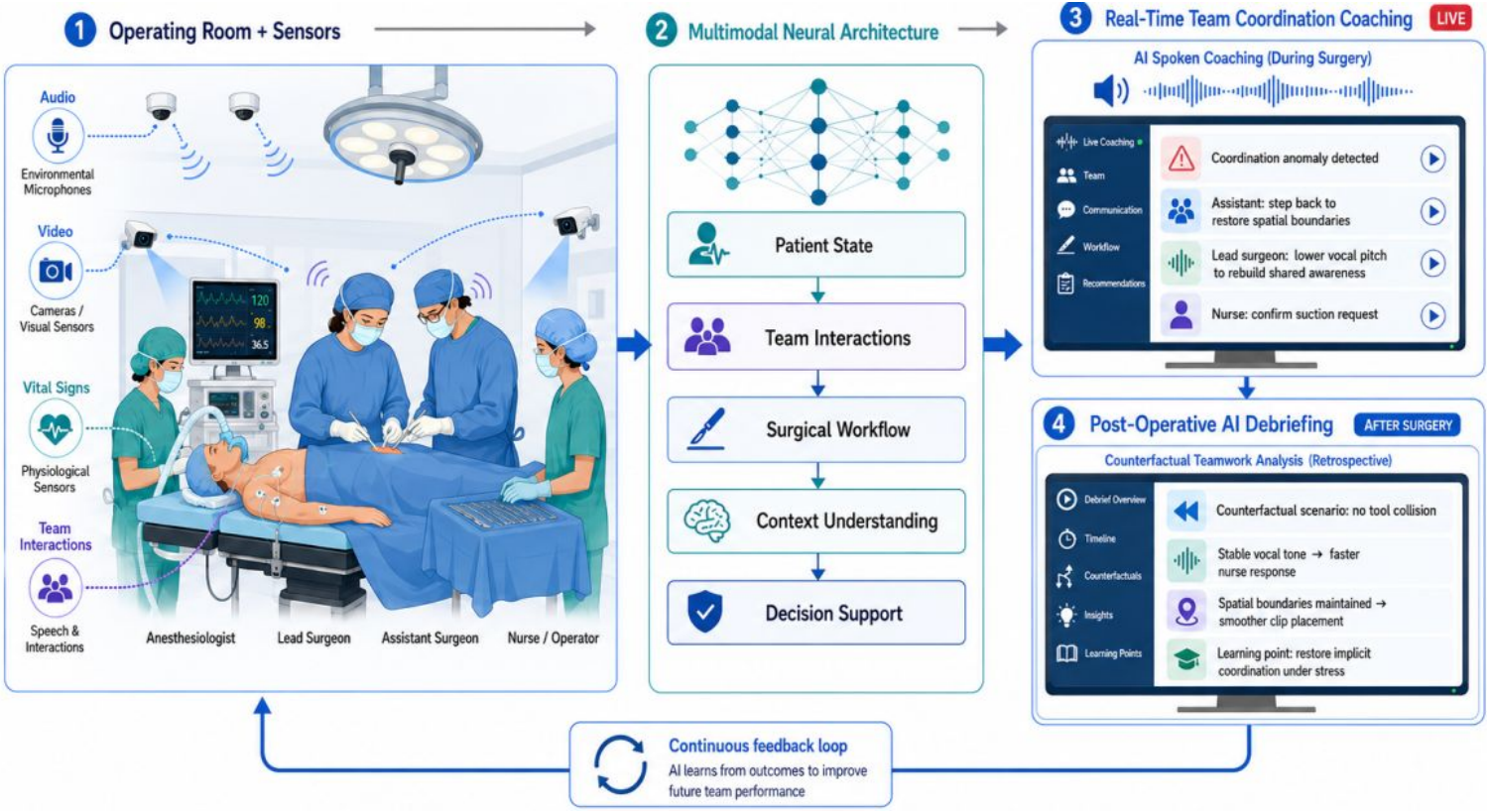
Operating Rooms Design



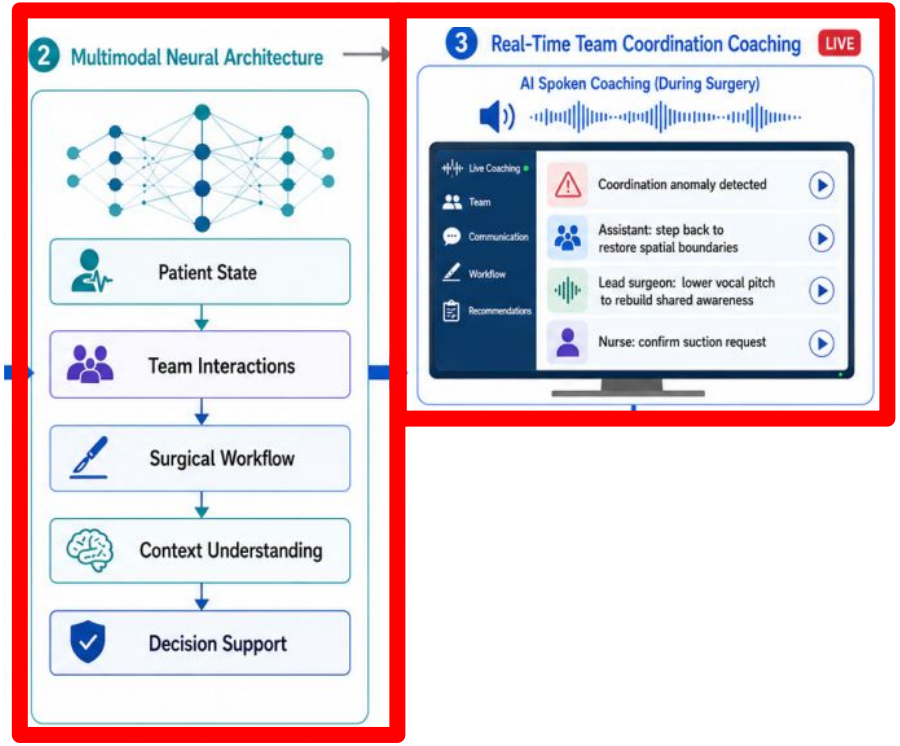
Operating Rooms Design



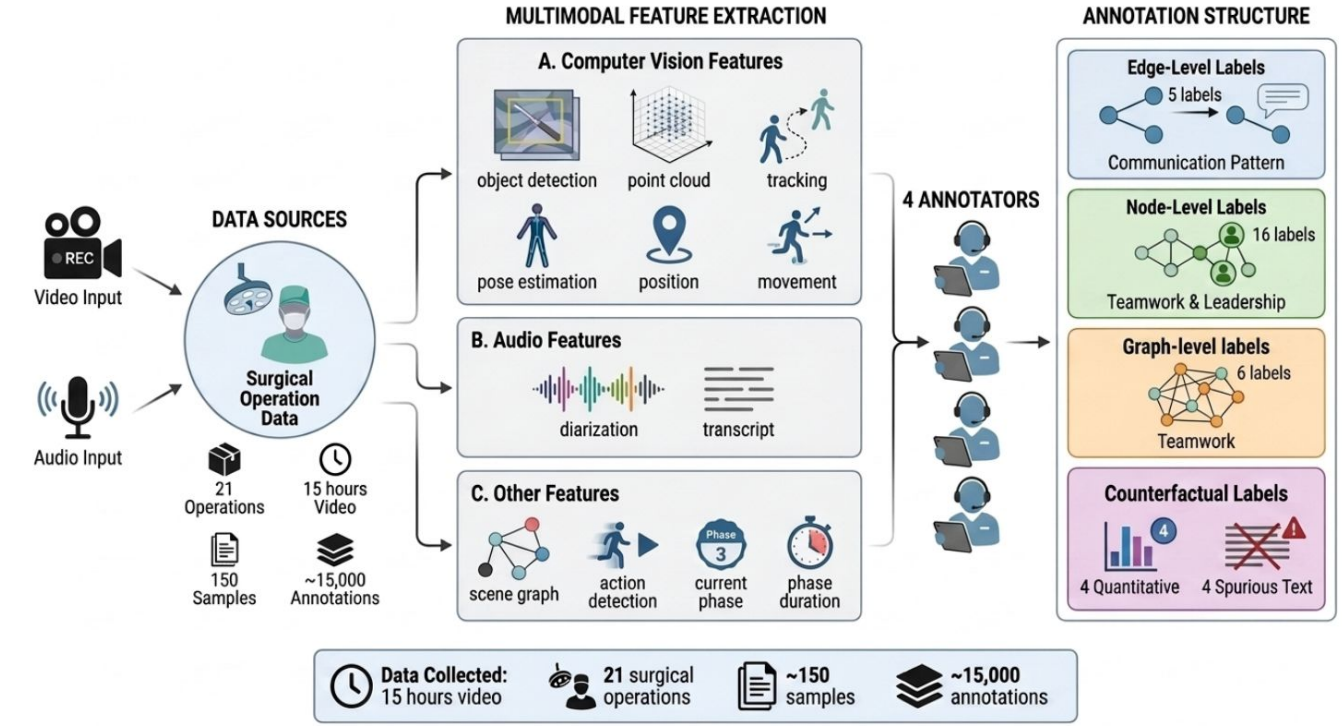
Operating Rooms Design



Operating Rooms Design



Operating Room Data



Feedback

FACTUAL EXPLAINABILITY

Why did the AI give this prediction?



AI PREDICTION:
HIGH RISK
of heart disease

WHY?

↑	High blood pressure (150 mmHg)	Strong impact
↑	High cholesterol (240 mg/dL)	Moderate impact
—	Age (45)	Low impact

It explains why the AI predicted high risk, based on **current data.**

VS

COUNTERFACTUAL EXPLAINABILITY

What could change the prediction?



AI PREDICTION:
HIGH RISK
of heart disease

WHAT COULD CHANGE THE PREDICTION?

If the following changes:

↓	Blood pressure from 150 to <130 mmHg	→	RISK BECOMES **LOW**
↓	Cholesterol from 240 to <190 mg/dL	→	RISK BECOMES **LOW**
🏃	Plus regular exercise (e.g., 30 min/day)	→	RISK BECOMES **LOW**

It shows what changes could lead to a **different (better) prediction.**

IN SHORT

Factual = explains the reason. **Counterfactual** = suggests what to change.



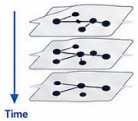
Model Operating Room

Operating room teams are inherently dynamic systems

A surgical procedure is not a static collaboration but a sequence of continuously evolving interactions among clinicians. Communication, coordination, and task allocation change according to the surgical phase, unexpected events, and workflow demands.

Why a temporal graph?

Static Network	Temporal Graph
Assumes relationships are constant	Captures interactions as they **evolve over time**
Ignores ordering of events	Preserves the **chronology** of communication
Misses transient collaborations	Reveals **short-lived** but critical interactions
Limited representation of workflow	Models the **dynamics** of teamwork throughout surgery



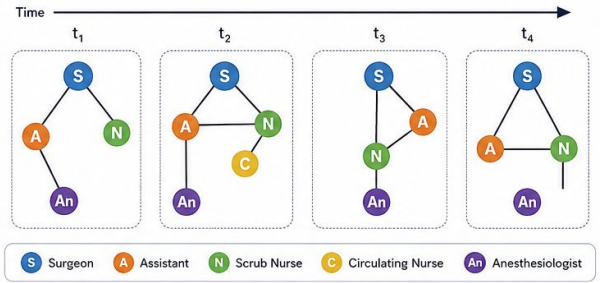
Scientific Motivation

- **Nodes:** members of the surgical team (surgeon, assistant, anesthesiologist, scrub nurse, circulating nurse, …)
- **Edges:** interactions (speech, gaze, instrument exchange, physical collaboration)
- **Time dimension:** every interaction is timestamped, creating a sequence of network snapshots

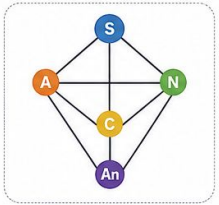
This representation enables the analysis of:

- Evolution of communication patterns
- Information flow during different surgical phases
- Emergence of leadership and coordination
- Detection of workflow disruptions
- Team adaptability during unexpected events

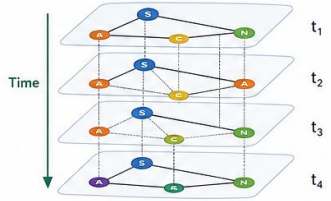
Interactions evolve over time



Static graph
(aggregated over the whole procedure)



Temporal graph
(sequence of time-respecting interactions)

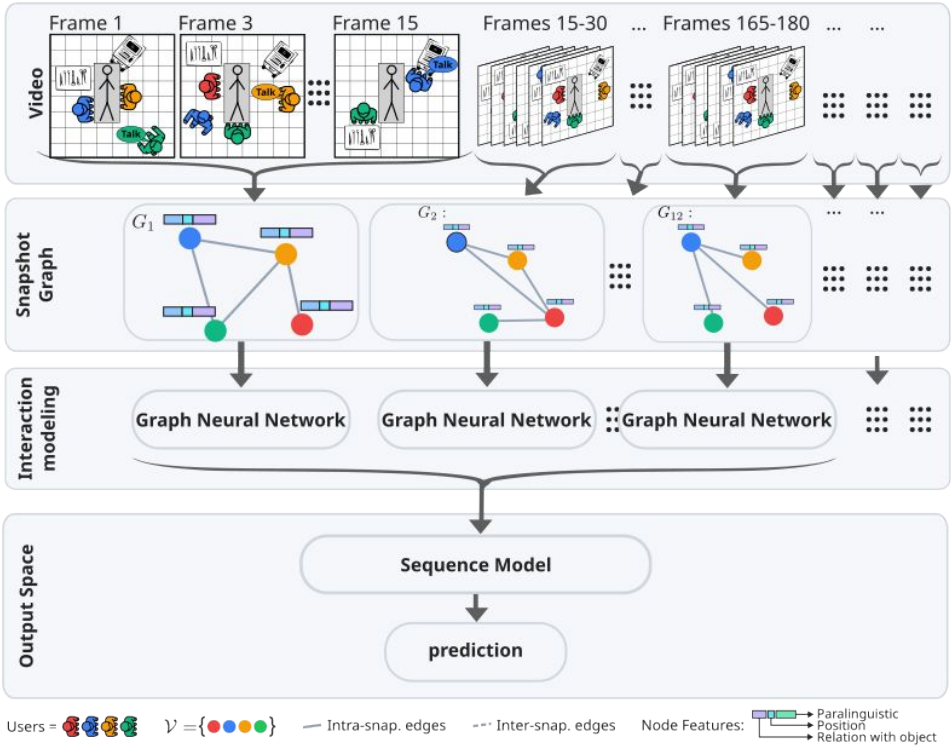


Loses temporal order
Does not capture when interactions occurred or in which sequence

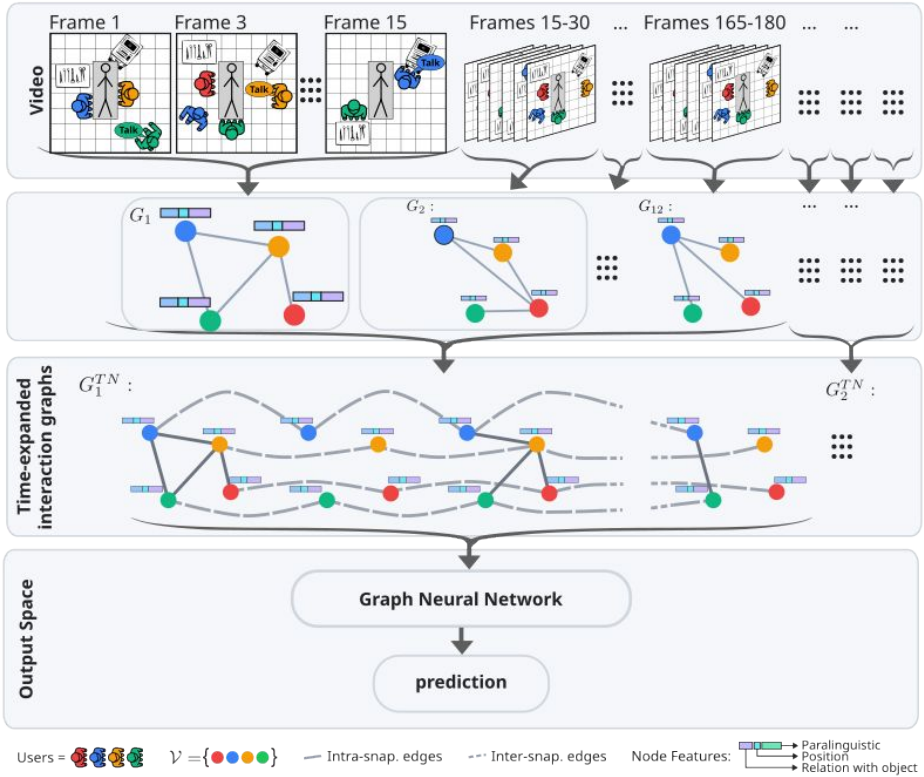
Preserves interaction dynamics
Captures who interacts with whom, when, and in what sequence

Key Insight: Temporal graphs preserve **who interacts with whom, when, and in what sequence,** providing a much richer representation of surgical teamwork than static networks.

Temporal Graph Neural Networks



Time-Expanded Graph Neural Networks

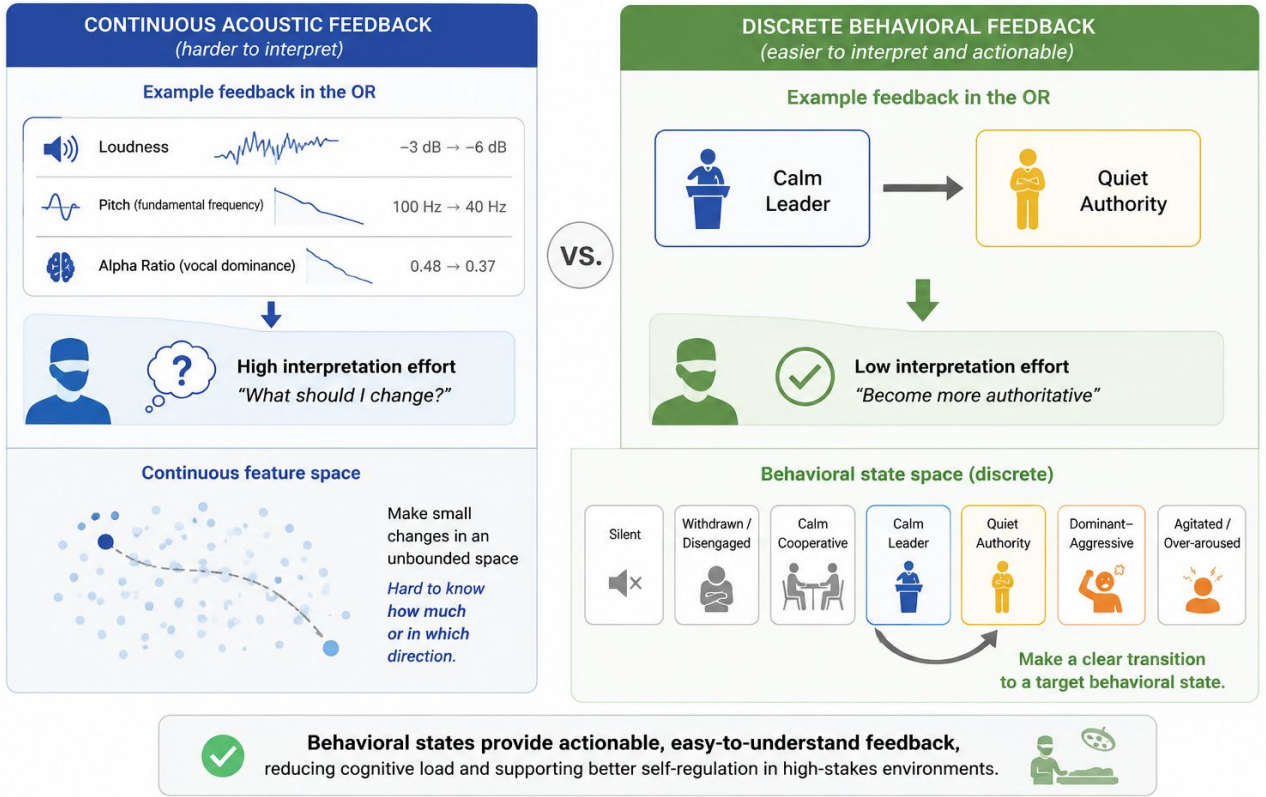


Time-Expanded performance

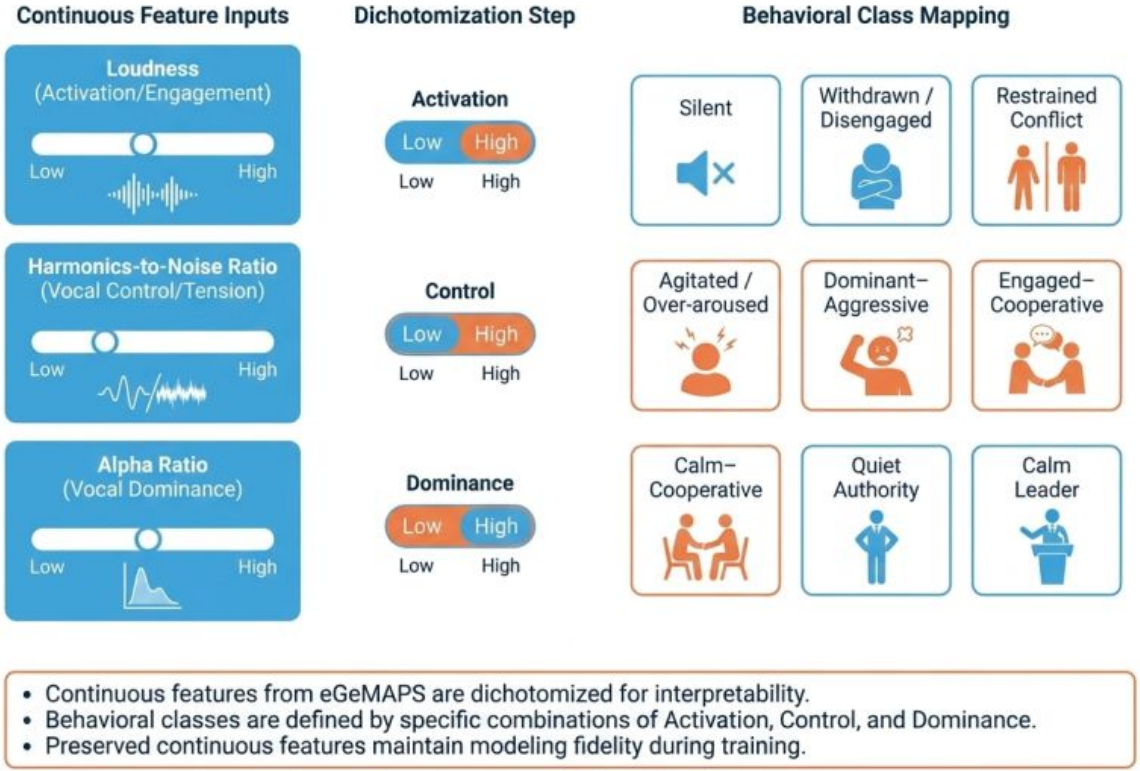
Paradigm	Model	Team		Individual				Interaction
		OTAS	Phase duration	NOTSS	BFTQ	SYMLOG	GLIS	HMT
SNN	MLP	59.8 ± 2.2	60.5 ± 1.2	51.5 ± 1.0	42.5 ± 1.1	48.5 ± 0.9	64.5 ± 0.8	41.5 ± 1.2
	RF	60.2 ± 0.8	61.8 ± 1.8	52.1 ± 1.2	38.5 ± 0.7	42.1 ± 0.6	65.2 ± 0.7	44.1 ± 1.0
TNN	LSTM	62.3 ± 1.6	62.8 ± 0.9	56.2 ± 1.4	52.2 ± 1.3	55.2 ± 0.9	64.8 ± 1.1	48.2 ± 1.1
	GRU	61.8 ± 1.8	63.0 ± 0.8	58.1 ± 1.1	55.5 ± 1.2	57.3 ± 0.8	62.6 ± 1.3	51.2 ± 1.3
	MHA	63.8 ± 2.0	62.4 ± 1.3	60.4 ± 1.2	56.2 ± 1.0	61.3 ± 0.8	64.9 ± 1.0	53.0 ± 1.2
RENN	GCN	61.4 ± 1.5	64.6 ± 0.6	58.2 ± 1.5	57.5 ± 1.1	59.8 ± 0.9	63.7 ± 1.2	56.2 ± 1.2
	GAT	62.1 ± 1.8	65.2 ± 0.8	59.3 ± 1.0	58.5 ± 1.2	59.6 ± 1.0	65.4 ± 1.2	58.0 ± 1.2
	GIN	65.2 ± 1.5	64.0 ± 1.2	61.2 ± 1.1	59.0 ± 1.3	60.4 ± 1.1	66.8 ± 1.0	58.3 ± 1.1
Space-and-Time TRENN	GCN+MHA	64.0 ± 1.5	67.3 ± 0.6	66.6 ± 1.1	62.8 ± 1.0	64.1 ± 0.8	62.8 ± 1.2	65.4 ± 1.6
	GAT+MHA	67.2 ± 1.7	67.1 ± 0.7	65.6 ± 1.1	65.8 ± 1.2	66.1 ± 1.1	62.0 ± 0.8	65.1 ± 1.2
	GIN+MHA	63.2 ± 1.5	65.2 ± 0.6	66.2 ± 1.0	65.2 ± 1.1	66.0 ± 1.2	64.2 ± 1.0	65.2 ± 1.3
Time-and-Space TRENN	MHA+GCN	65.2 ± 1.3	64.8 ± 0.8	64.0 ± 1.2	65.2 ± 1.8	63.8 ± 1.0	65.2 ± 0.9	69.4 ± 1.1
	MHA+GAT	65.8 ± 1.5	64.8 ± 1.0	64.1 ± 1.0	65.1 ± 1.6	64.0 ± 1.2	67.6 ± 0.9	66.8 ± 1.1
	MHA+GIN	64.2 ± 1.2	65.6 ± 0.9	65.0 ± 1.2	64.1 ± 1.4	63.2 ± 1.1	68.4 ± 1.0	67.9 ± 1.0
TE-RENN	TE-GCN	72.4 ± 1.7	<u>70.1 ± 1.0</u>	71.0 ± 1.3	76.1 ± 1.4	70.2 ± 1.1	75.4 ± 1.0	73.9 ± 1.2
	TE-GAT	<u>73.1 ± 1.7</u>	69.8 ± 0.8	71.7 ± 1.2	75.5 ± 1.1	71.5 ± 1.1	74.8 ± 1.0	**73.9 ± 0.8**
	TE-GIN	**74.2 ± 1.7**	**72.5 ± 1.0**	**72.5 ± 1.0**	76.7 ± 1.1	73.9 ± 1.0	**76.2 ± 1.2**	73.5 ± 0.8

Table 6: Combined results measuring F1-macro according to a LOGO protocol, with experiments runned over 10 different random seeds. Tasks are organized into three levels: Team (OTAS, Phase duration), Individual (NOTSS, BFTQ, SYMLOG, GLIS), and Interaction (HMT). Bold indicates best, underline indicates second best.

Interpretable feedback



Interpretable feedback



- Continuous features from eGeMAPS are dichotomized for interpretability.
- Behavioral classes are defined by specific combinations of Activation, Control, and Dominance.
- Preserved continuous features maintain modeling fidelity during training.

Feedback example

1) Adding Communication



Situation. The nurse asks a question, but the leader is distracted and does not respond, leading to silence and low communication.

Counterfactual. Add edges

Interpretation. Reduce disengagement by increasing leader responsiveness.

2) Reducing Off-Topic Talk

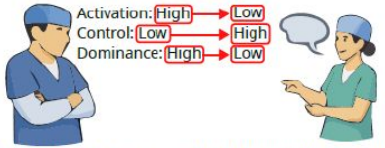


Situation. Irrelevant discussions introduce excessive interactions, distracting the team and lowering situational awareness.

Counterfactual. Remove edges

Interpretation. Reduce irrelevant communication.

3) Lack of Responsiveness



Situation. A nurse asks for help, but the leader does not respond, affecting team cooperation.

Counterfactual. Decrease "Activation" and "Dominance", increase "Control".

Interpretation. Adopt a more composed and responsive communication style ("Calm–Cooperative").

4) Aggressive Communication



Situation. A rude response from the leader to a nurse's question reduces team cooperation.

Counterfactual. Increase "Control" (low → high).

Interpretation. Adopt a "Calm Leader" style instead of "Dominant–Aggressive".

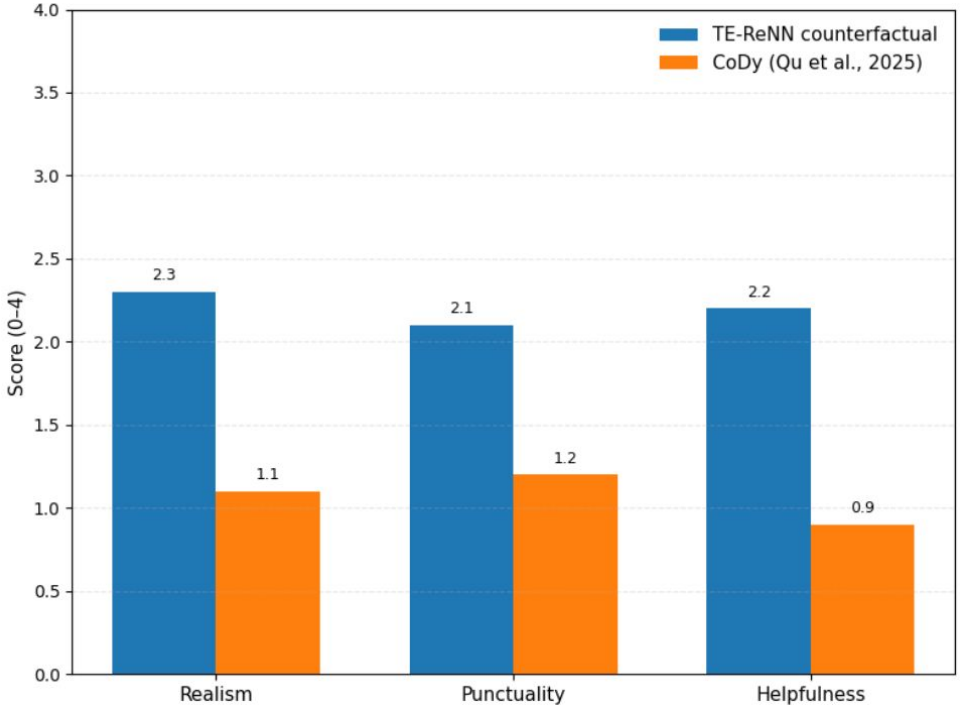
Counterfactual algorithm

Algorithm 1 Actionable Counterfactual Generation

Require: Input prefix $\mathcal{G}_{1:t}$, target outcome y^{target}, encoder f_θ, repository $\mathcal{Z}$
Ensure: Actionable counterfactual explanation e

Offline: Build repository $\mathcal{Z}$
for each training sequence $\mathcal{G}^i_{1:T_i}$ **do**
  $z^i \leftarrow f_\theta(\mathcal{G}^i_{1:T_i})$
end for
Online: Encoding
$\mathbf{z} \leftarrow f_\theta(\mathcal{G}_{1:t})$
Retrieval of the most similar sample
$(\mathcal{G}^*_{1:t}, y^{\text{retrieved}}) \leftarrow \textsc{Retrieve}(\mathbf{z}, \mathcal{Z}, y^{\text{target}})$
Factual explanation of target samples
$\mathcal{E} \leftarrow \textsc{Explain}(\mathcal{G}^*_{1:t}, f_\theta)$
Selection
for each explanation $e \in \mathcal{E}$ (ranked by importance) **do**
  if $\textsc{Match}(e, \mathcal{G}_{1:t}) =$ **false then**
    return e
  end if
end for
return $\emptyset$

Counterfactual assessment



Thank you for the attention!

Appendix - Behavioral classes

Activation	Control	Dominance	Behavioral Class
Low	Low	Low	Withdrawn / Disengaged
Low	Low	High	Restrained Conflict
Low	High	Low	Calm–Cooperative
Low	High	High	Quiet Authority
High	Low	Low	Agitated / Over-aroused
High	Low	High	Dominant–Aggressive
High	High	Low	Engaged–Cooperative
High	High	High	Calm Leader

Table 1: Behavioral classes defined by combining three paralinguistic features extracted from the eGeMAPS set.